Research Methodology

Week 2 Assignment: Questions, Answers and Explanations

Week 2: Literature Survey and Experimental Skills

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

Correct answers are the NPTEL portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. This week covers why and how to survey literature, types of sources and citation details.

Answer key at a glance

Question	Type	Correct option(s)	Your portal score
1	Single answer	**C**	1 (full marks)
2	Select all that apply	**A, B, C, D**	0.25 (partial: not all four correct options chosen)
3	Single answer	**D**	1 (full marks)
4	Single answer	**B**	1 (full marks)
5	Single answer	**B**	1 (full marks)
6	Single answer	**B**	1 (full marks)
7	Single answer	**B**	1 (full marks)
8	Select all that apply	**A, B**	0.5 (partial: only one of the two correct options chosen)
9	Select all that apply	**A, B, D**	0.33 (partial: only one of the three correct options chosen)
10	Single answer	**B**	1 (full marks)

Questions with explanations

Question 1 of 10

Experimental data generated from a setup must validate the theoretical predictions or models designed for that setup.

Single answer.

Option	Statement	Verdict	Reason
A	True	**Wrong**	"Must" is too strong: data can disagree with a model because of experimental error, model limits or a new effect.
B	False	**Wrong**	Data often do agree with theory, so "always false" is wrong too.
C	**Sometimes**	**Correct**	Data confirm the theory in some cases and not in others, and a mismatch is itself a valid result.
D	Never	**Wrong**	Many experiments do confirm their predictions.

Correct answer: C.

Explanation: Experiments are used to test a theory, not to force agreement with it. When the data match, the model is supported; when they do not, the cause may be an error in the setup, a limit of the model or a genuinely new effect. So validation happens only sometimes.

Question 2 of 10

Literature survey is important because

Select all that apply. *Your portal score: 0.25 (partial: not all four correct options chosen).*

Option	Statement	Verdict	Reason
A	**It helps in getting a summary of work that has been done till date in some particular research area**	**Correct**	A survey tells you what is already known in the area.
B	**It is helpful in ensuring that a research work is not getting repeated**	**Correct**	Knowing existing work stops you from duplicating it.
C	**It is helpful in understanding the approaches taken by scientists and what are the gaps in the processes**	**Correct**	It shows the methods used and the gaps you can fill.
D	**It is helpful in clarifying the controversial results that one can get**	**Correct**	Earlier findings help you judge conflicting or surprising results.

Correct answer: A and B and C and D.

Explanation: A literature survey gives a summary of existing work, prevents repetition, shows the approaches used and the gaps that remain, and helps you interpret controversial results. All four statements hold.

Question 3 of 10

Which of the following methods of searching for literature is suitable to identify the earliest contributor of a scientific outcome?

Single answer.

Option	Statement	Verdict	Reason
A	Keyword based search followed by sorting by year of publication	**Wrong**	This only gives the oldest paper that matches your keywords, which need not be the origin of the idea.
B	Citation based search followed by sorting by year of publication	**Wrong**	A citation search moves forward in time to later papers, not back to the origin.
C	Keyword based search followed by sorting by citations	**Wrong**	Sorting by citations finds the most influential papers, not the earliest.
D	**Back reference based search following a keyword based search**	**Correct**	Start from a paper found by keyword, then follow its reference list backwards until you reach the original source.

Correct answer: D.

Explanation: Each paper cites the work it builds on. By repeatedly reading the reference lists backwards (back-referencing) you reach the first paper that reported the outcome.

Question 4 of 10

Which of the following bibliographic details is a sufficient and unique piece of information to locate an article in an archival journal?

Single answer.

Option	Statement	Verdict	Reason
A	ISSN	**Wrong**	The ISSN identifies the whole journal, not one article.
B	**DOI**	**Correct**	A digital object identifier is assigned to a single article and resolves directly to it.
C	Page Number	**Wrong**	A page number means something only together with the journal, volume and year.
D	Volume Number	**Wrong**	A volume contains many articles, so it is not unique.

Correct answer: B.

Explanation: The DOI is a permanent, unique identifier for one article. The other details identify the journal or a position inside it and need other details alongside them.

Question 5 of 10

Which of the following is always true for an open access journal?

Single answer.

Option	Statement	Verdict	Reason
A	Such journals are not peer reviewed	**Wrong**	Many open access journals are rigorously peer reviewed.
B	**Articles in such a journal are freely available for download**	**Correct**	Free access for readers is the definition of open access.
C	Such journals charge for publication of an article	**Wrong**	Some charge authors a fee and some do not, so it is not always true.
D	Such journals are of dubious reputation	**Wrong**	Many open access journals are well respected.

Correct answer: B.

Explanation: Open access means that readers can download the articles free of charge. Peer review, author fees and reputation vary from journal to journal, so only free availability is always true.

Question 6 of 10

Which of the following is NOT a recommended practice to identify publications that need to be listed under references of an article?

Single answer.

Option	Statement	Verdict	Reason
A	Relevant publications actually referred to	**Wrong**	Citing the works you actually used is recommended, so it is not the answer.
B	**Latest publications by the editor of the journal**	**Correct**	Citing the editor's papers just because of who the editor is does not help the reader and looks like padding.
C	Original publications in the topic of the article	**Wrong**	Citing the original sources of an idea is recommended.
D	Own publications referred in the article for further details	**Wrong**	Citing your own related work for further details is acceptable.

Correct answer: B.

Explanation: References should be the works you actually used, the original sources of the ideas and your own related work where the reader needs it. Adding an editor's recent papers only because of their position is not a sound reason to cite.

Note: *This question asks for the practice that is NOT recommended, so "Correct" marks the bad practice that you should select.*

Question 7 of 10

Which of the following indices is considered as a measure of the impact of a particular author in its area of his/her specialization?

Single answer.

Option	Statement	Verdict	Reason
A	Total number of articles published	**Wrong**	A count of papers says nothing about how much they were cited.
B	**Hirsch index of the author**	**Correct**	The h-index is the largest h such that the author has h papers each cited at least h times, so it combines output and citations.
C	Number of views of his/her latest article	**Wrong**	Views of one article do not show the author's overall impact.
D	Number of downloads of all articles in a year	**Wrong**	Downloads measure interest in a year, not lasting influence.

Correct answer: B.

Explanation: The Hirsch index measures both productivity and impact. For example, an author with 5 papers each cited at least 5 times (and no more than 5 such papers) has h = 5.

Question 8 of 10

The following are categorized as primary source for literature survey

Select all that apply. *Your portal score: 0.5 (partial: only one of the two correct options chosen).*

Option	Statement	Verdict	Reason
A	**Periodicals & research reports**	**Correct**	They report original research first-hand.
B	**Conference proceedings & official publications**	**Correct**	They are original, first-hand reports of work and data.
C	Compilations & reviews	**Wrong**	They summarise other people's work, so they are secondary sources.
D	Directories & year books	**Wrong**	They are reference tools for locating information (tertiary sources).

Correct answer: A and B.

Explanation: A primary source presents original findings. Journal articles, research reports, conference papers and official publications fit this. Reviews and compilations are secondary, and directories and year books are reference aids.

Question 9 of 10

The abstracting databases for systematic manual searches include

Select all that apply. *Your portal score: 0.33 (partial: only one of the three correct options chosen).*

Option	Statement	Verdict	Reason
A	**INSPEC and COMPENDEX**	**Correct**	INSPEC covers physics, electrical and computing literature; COMPENDEX covers engineering.
B	**PubMed**	**Correct**	PubMed is the abstracting and indexing database for biomedical literature.
C	Wikipedia	**Wrong**	Wikipedia is an open encyclopedia, not an abstracting database.
D	**Chemical Abstract (CA) service**	**Correct**	Chemical Abstracts indexes and abstracts chemistry literature.

Correct answer: A and B and D.

Explanation: Abstracting databases index journal articles with short abstracts so that you can search them systematically. INSPEC, COMPENDEX, PubMed and Chemical Abstracts are such services; Wikipedia is not.

Question 10 of 10

The six resources collected

(1) A reference in a journal article that you have not referred yet. (2) A link to the website of a professional society. (3) A published article with journal, year, volume and page number. (4) An article from a newly launched journal with volume number and page number, but year and ISSN are due. (5) A newly accepted journal article with a DOI link. (6) A book with an ISBN number available.

Which of the following can be considered as archival (valuable record) references for inclusion in a publication?

Single answer.

Option	Statement	Verdict	Reason
A	Only 1, 2 and 3	**Wrong**	Item 1 is unread and unverified, and item 2 is a website that can change or disappear.
B	**Only 3, 5 and 6**	**Correct**	Item 3 has a full citation, item 5 has a permanent DOI and item 6 is a published book with an ISBN.
C	Only 3, 4 and 6	**Wrong**	Item 4 lacks its year and ISSN, so it cannot yet be traced reliably.
D	Only 1, 2 and 5	**Wrong**	Items 1 and 2 are not archival, as above.

Correct answer: B.

Explanation: An archival reference is permanent and traceable. A complete journal citation (3), a DOI (5) and a book with an ISBN (6) qualify. An unread reference (1), a website (2) and an incompletely identified article (4) do not.